МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ

НАЦІОНАЛЬНИЙ УНІВЕРСИТЕТ “ЛЬВІВСЬКА ПОЛІТЕХНІКА”

Кафедра систем штучного інтелекту



Лабораторна робота № 5

з дисципліни «Хмарні технології»

Виконав:

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Лозовий Олег

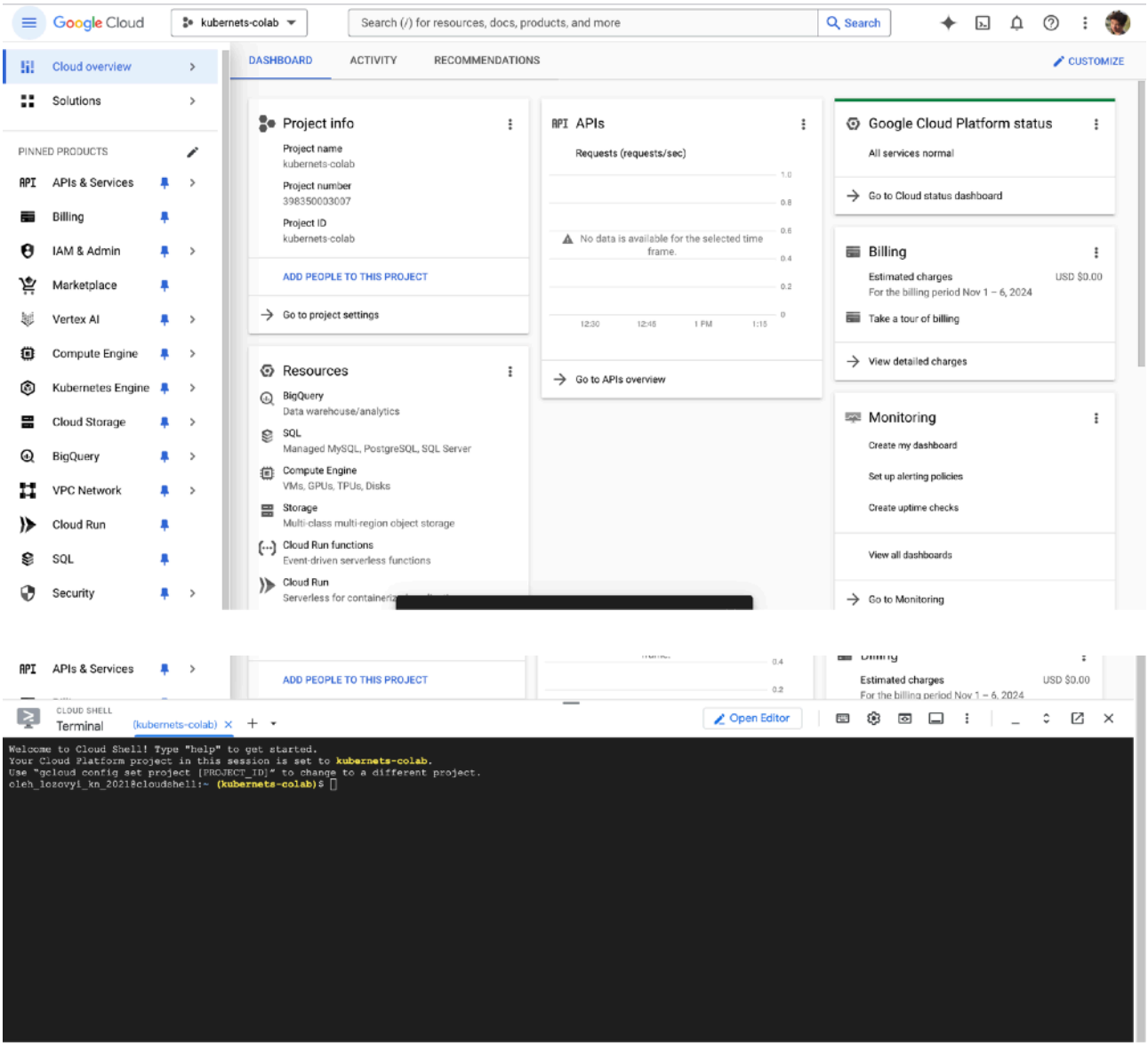
Викладач:

Гасько Р.Т.

Львів – 2024р.

Тема: Microservices on Google Kubernetes.

Хід роботи:




```
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to kubernets-colab.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ gcloud auth list
Credentialed Accounts

ACTIVE: *
ACCOUNT: oleh.lozovyi.kn.2021@lpnu.ua

To set the active account, run:
    $ gcloud config set account `ACCOUNT`
```

```
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ gcloud config list project
[core]
project = kubernets-colab

Your active configuration is: [cloudshell-18683]
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$
```

```
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ gcloud config set project kubernets-colab
Updated property [core/project].
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$
```

```
kubernets-colab
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ gcloud config set compute/zone us-central1-f
API [compute.googleapis.com] not enabled on project [kubernets-colab]. Would you like to enable and retry (this will take a few minutes)? (y/N)?  Y

Enabling service [compute.googleapis.com] on project [kubernets-colab]...



Operation "operations/acf.p2-398350003007-c9eb620a-5ac3-4762-bc21-6543d2169a50" finished successfully.
WARNING: Property validation for compute/zone was skipped.
Updated property [compute/zone].
```

```
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ cd ~
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ git clone https://github.com/googlecodelabs/monolith-to-microservices.git
Cloning into 'monolith-to-microservices'...
remote: Enumerating objects: 1201, done.
remote: Counting objects: 100% (306/306), done.
remote: Compressing objects: 100% (155/155), done.
remote: Total 1201 (delta 234), reused 189 (delta 146), pack-reused 895 (from 1)
Receiving objects: 100% (1201/1201), 3.82 MiB | 9.95 MiB/s, done.
Resolving deltas: 100% (583/583), done.
oleh_lozovyi_kn_2021@cloudshell:~ (kubernets-colab)$ cd ~/monolith-to-microservices
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ ./setup.sh
Installing monolith dependencies...

added 68 packages, and audited 69 packages in 3s

14 packages are looking for funding
  run `npm fund` for details

4 vulnerabilities (1 low, 3 moderate)

To address all issues, run:
  npm audit fix

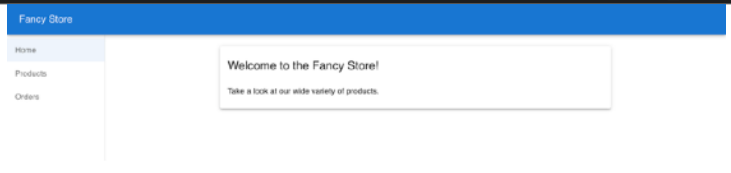
Run `npm audit` for details.
Completed
```

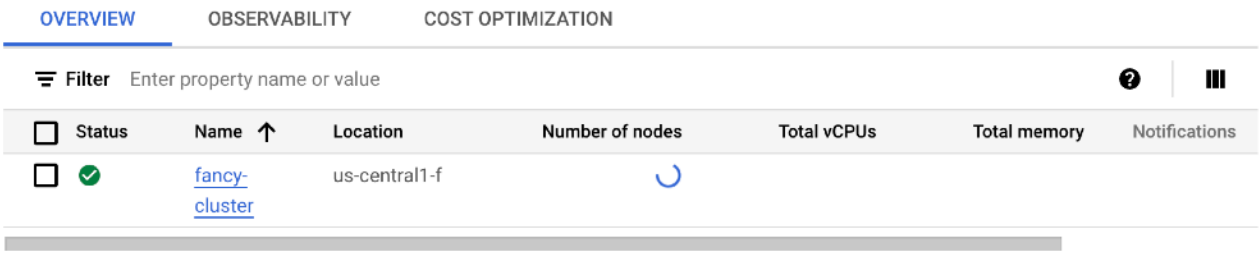
```
Setup completed successfully!
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ gcloud services enable container.googleapis.com
Operation "operations/acf.p2-398350003007-b6Bb4bb9-8eb3-4c74-a0a6-9b04bcf98565" finished successfully.
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ gcloud container clusters create fancy-cluster --num-nodes 3
Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. See https://cloud.google.com/kubernetes-engine/docs/how-to
/disable-kubelet-readonly-port for ways to check usage and for migration instructions.
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster fancy-cluster in us-central1-f... Cluster is being health-checked (Kubernetes Control Plane is healthy)...working.
                                                                                                                                                                Crea
Creating cluster fancy-cluster in us-central1-f... Cluster is being health-checked (Kubernetes Control Plane is healthy)...working.
                                                                                                                                                               Creating
Creating cluster fancy-cluster in us-central1-f... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.
Created [https://container.googleapis.com/v1/projects/kubernets-colab/zones/us-central1-f/clusters/fancy-cluster].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload_/gcloud/us-central1-f/fancy-cluster?project=kubernets-colab
kubeconfig entry generated for fancy-cluster.
NAME: fancy-cluster
LOCATION: us-central1-f
MASTER_VERSION: 1.30.5-gke.1355000
MASTER_IP: 34.171.54.220
MACHINE_TYPE: e2-medium
NODE_VERSION: 1.30.5-gke.1355000
NUM_NODES: 3
STATUS: RUNNING
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$
```

```
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ gcloud compute instances list
NAME: gke-fancy-cluster-default-pool-cfb5bcdf-152g
ZONE: us-central1-f
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.3
EXTERNAL_IP: 34.170.235.124
STATUS: RUNNING

NAME: gke-fancy-cluster-default-pool-cfb5bcdf-91kw
ZONE: us-central1-f
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.5
EXTERNAL_IP: 34.170.223.17
STATUS: RUNNING

NAME: gke-fancy-cluster-default-pool-cfb5bcdf-hshm
ZONE: us-central1-f
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.4
EXTERNAL_IP: 34.123.101.135
STATUS: RUNNING
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$
```






```
EXTERNAL_IP: 34.123.101.133
STATUS: RUNNING
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ cd ~/monolith-to-microservices
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ ./deploy-monolith.sh
Enabling Cloud Build APIs...
Operation "operations/acf.p2-398350003007-2c5ccca8-b4bf-470c-91cd-7b139c77f57d" finished successfully.
Completed.

Building Monolith Container...
Creating temporary archive of 27 file(s) totalling 2.4 MiB before compression.
Uploading tarball of [.] to [gs://kubernets-colab_cloudbuild/source/1730893689.120783-e1c6e53d28cd43b8b5ddb2bf7fa940fc.tgz]
ERROR: (gcloud.builds.submit) FAILED_PRECONDITION: invalid bucket "398350003007.cloudbuild-logs.googleusercontent.com"; servic
com does not have access to the bucket
Completed.

Deploying Monolith To GKE Cluster...
deployment.apps/monolith created
service/monolith exposed
Completed.

Please run the following command to find the IP address for the monolith service: kubectl get service monolith

Deployment completed successfully!
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ kubectl get service monolith
NAME        TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
monolith    LoadBalancer   34.118.235.143   <pending>     80:31582/TCP   6s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$
```

```
d41bcd3a037b: Preparing
fe0d845e767b: Preparing
f25ec1d93a58: Preparing
794ce8b1b516: Preparing
3220beed9b06: Preparing
684f82921421: Preparing
9af5f53e8f62: Preparing
fe0d845e767b: Waiting
f25ec1d93a58: Waiting
794ce8b1b516: Waiting
3220beed9b06: Waiting
9af5f53e8f62: Waiting
684f82921421: Waiting
be322b479aee: Layer already exists
d41bcd3a037b: Layer already exists
ec12fd85231f: Pushed
99653838b1f5: Pushed
fe0d845e767b: Layer already exists
13dd48dba143: Pushed
f25ec1d93a58: Layer already exists
c6a7fb432b52: Pushed
794ce8b1b516: Layer already exists
3220beed9b06: Layer already exists
684f82921421: Layer already exists
9af5f53e8f62: Layer already exists
1.0.0: digest: sha256:7aaa562a41db7072fcfe829bff2c1fa168039958fbe475c57eee75f16e528695 size: 2841
DONE
--------------------------------------------------------------------------------------------------------------------------
ID: f4cd3aa9-5289-444d-9fbb-61e62161dda1
CREATE_TIME: 2024-11-06T11:56:22+00:00
DURATION: 53S
SOURCE: gs://kubernets-colab_cloudbuild/source/1730894179.770163-8ef1e514e2f34c0790eaf7ed9893ee9d.tgz
IMAGES: gcr.io/kubernets-colab/monolith:1.0.0
STATUS: SUCCESS
Completed.

Deploying Monolith To GKE Cluster...
error: failed to create deployment: deployments.apps "monolith" already exists
Error from server (AlreadyExists): services "monolith" already exists
Completed.

Please run the following command to find the IP address for the monolith service: kubectl get service monolit

Deployment completed successfully!
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$ kubectl get service monolith
NAME        TYPE           CLUSTER-IP      EXTERNAL-IP     PORT(S)        AGE
monolith    LoadBalancer   34.118.235.143   34.68.107.245   80:31582/TCP   9m7s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (kubernets-colab)$
```

```
Deployment completed successfully!
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices (manibba)$ cd ~/monolith-to-microservices/microservices/src/orders
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ gcloud builds submit --tag gcr.io/${GOOGLE_CLOUD_PROJECT}/orders:1.0.0 .
Creating temporary archive of 8 file(s) totalling 24.3 KiB before compression.
Some files were not included in the source upload.

Check the gcloud log [/tmp/tmp.isvDQtxfMi/logs/2024.11.06/14.14.24.835982.log] to see which files and the contents of the
default gcloudignore file used (see `$ gcloud topic gcloudignore` to learn
more).

Uploading tarball of [.] to [gs://manibba_cloudbuild/source/1730902465.122618-72d46af0980745f59e781695abc6c0ca.tgz]
Created [https://cloudbuild.googleapis.com/v1/projects/manibba/locations/global/builds/a19bf647-d958-403e-8613-4525df395e5f].
Logs are available at [ https://console.cloud.google.com/cloud-build/builds/a19bf647-d958-403e-8613-4525df395e5f?project=1049491440578 ].
Waiting for build to complete. Polling interval: 1 second(s).
------------------------------------------------------------------------ REMOTE BUILD OUTPUT ------------------------------------------------------------------------
starting build "a19bf647-d958-403e-8613-4525df395e5f"

FETCHSOURCE
Fetching storage object: gs://manibba_cloudbuild/source/1730902465.122618-72d46af0980745f59e781695abc6c0ca.tgz#1730902465853016
Copying gs://manibba_cloudbuild/source/1730902465.122618-72d46af0980745f59e781695abc6c0ca.tgz#1730902465853016...
/ [1 files][  8.9 KiB/  8.9 KiB]
Operation completed over 1 objects/8.9 KiB.
BUILD
Already have image (with digest): gcr.io/cloud-builders/docker
```

```
STATUS: SUCCESS
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl create deployment orders --image=gcr.io/${GOOGLE_CLOUD_PROJECT}/orders:1.0.0
deployment.apps/orders created
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get all
NAME                          READY   STATUS    RESTARTS   AGE
pod/monolith-79f9b5cdf9-9zqrq   1/1     Running   0          9m33s
pod/orders-6f8f99c944-c7961     1/1     Running   0          6s

NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP    PORT(S)        AGE
service/kubernetes   ClusterIP      34.118.224.1     <none>         443/TCP        12m
service/monolith     LoadBalancer   34.118.233.232   34.118.47.58   80:31063/TCP   9m32s

NAME                       READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/monolith   1/1     1            1           9m33s
deployment.apps/orders     1/1     1            1           7s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/monolith-79f9b5cdf9   1         1         1       9m33s
replicaset.apps/orders-6f8f99c944     1         1         1       7s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$
```

```
replicaset.apps/orders-6f8f99c944     1         1         1       7s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl expose deployment orders --type=LoadBalancer --port 80 --target-port 8081
service/orders exposed
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get service orders
NAME     TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
orders   LoadBalancer   34.118.231.235   <pending>     80:32483/TCP   5s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get service orders
NAME     TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
orders   LoadBalancer   34.118.231.235   <pending>     80:32483/TCP   15s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get service orders
NAME     TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
orders   LoadBalancer   34.118.231.235   <pending>     80:32483/TCP   18s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get service orders
NAME     TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
orders   LoadBalancer   34.118.231.235   <pending>     80:32483/TCP   29s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$ kubectl get service orders
NAME     TYPE           CLUSTER-IP       EXTERNAL-IP     PORT(S)        AGE
orders   LoadBalancer   34.118.231.235   34.118.29.240   80:32483/TCP   45s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/orders (manibba)$
```

```
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/react-app (manibba)$ npm run build:monolith

> frontend@0.1.0 build:monolith
> env-cmd -f .env.monolith react-scripts build

Creating an optimized production build...
Compiled successfully.

File sizes after gzip:

  88.22 kB (+7 B)  build/static/js/main.97f4fbf7.js

The project was built assuming it is hosted at /.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.
You may serve it with a static server:

  npm install -g serve
  serve -s build

Find out more about deployment here:

  https://cra.link/deployment


> frontend@0.1.0 postbuild:monolith
> node scripts/post-build.js ./build ../monolith/public

Deleting stale folder: ../monolith/public
Deleted stale destination folder: ../monolith/public
Copying files from ./build to ../monolith/public
Copied ./build to ../monolith/public successfully!
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/react-app (manibba)$ cd ~/monolith-to-microservices/monolith
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/monolith (manibba)$ gcloud builds submit --tag gcr.io/${GOOGLE_CLOUD_PROJECT}/monolith:2.0.0 .
```

```
ID: 57a6cb92-86f6-4271-a3d7-43cf5578b617
CREATE_TIME: 2024-11-06T14:19:43+00:00
DURATION: 29S
SOURCE: gs://manibba_cloudbuild/source/1730902780.977388-b573b1ba5bad49899e30f0a8afdd984e.tgz
IMAGES: gcr.io/manibba/monolith:2.0.0
STATUS: SUCCESS
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/monolith (manibba)$ kubectl set image deployment/monolith monolith=gcr.io/${GOOGLE_CLOUD_PROJECT}/monolith:2.0.0
deployment.apps/monolith image updated
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/monolith (manibba)$
```

global (non-regional)

Filter Enter property name or value

Status	Build	Source	Ref	Commit	Trigger Name	Created	Duration
✓	3c2ea7b1	Googl...	-	-	-	11/6/24, 3:05 PM	39 sec
✓	49b54750	Googl...	-	-	-	11/6/24, 3:02 PM	1 min 4 sec

Orders

Order Id	Date	Total Items	Cost
ORD-000001-MICROSERVICE	7/01/2019	1	$67.99
ORD-000002-MICROSERVICE	7/24/2019	1	$124
ORD-000003-MICROSERVICE	8/03/2019	1	$12.49
ORD-000004-MICROSERVICE	8/14/2019	2	$89.83
ORD-000005-MICROSERVICE	8/29/2019	1	$12.3

```
ID: 176fbb9e-da60-4907-b0a1-c1ed68b1eae7
CREATE_TIME: 2024-11-06T14:21:36+00:00
DURATION: 37S
SOURCE: gs://manibba_cloudbuild/source/1730902895.576213-8865f566c5c446138abadcbfcffb4ab0.tgz
IMAGES: gcr.io/manibba/products:1.0.0
STATUS: SUCCESS
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl create deployment products --image=gcr.io/${GOOGLE_CLOUD_PROJECT}/products:1.0.0
deployment.apps/products created
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$
```

```
deployment.apps/products created
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl expose deployment products --type=LoadBalancer --port 80 --target-port 8082
service/products exposed
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl get service products
NAME       TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
products   LoadBalancer   34.118.235.218   <pending>     80:30281/TCP   5s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl get service products
NAME       TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
products   LoadBalancer   34.118.235.218   <pending>     80:30281/TCP   12s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl get service products
NAME       TYPE           CLUSTER-IP       EXTERNAL-IP      PORT(S)        AGE
products   LoadBalancer   34.118.235.218   34.118.17.238    80:30281/TCP   46s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$
```

```
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl get
NAME       TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
products   LoadBalancer   34.118.235.218   <pending>     80:30281/TCP   12s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ kubectl get
NAME       TYPE           CLUSTER-IP       EXTERNAL-IP      PORT(S)        AGE
products   LoadBalancer   34.118.235.218   34.118.17.238    80:30281/TCP   46s
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/microservices/src/products (manibba)$ cd ~/monoli
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/react-app (manibba)$ nano .env.monolith
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/react-app (manibba)$ nano .env.monolith
oleh_lozovyi_kn_2021@cloudshell:~/monolith-to-microservices/react-app (manibba)$ npm run build:monolith

> frontend@0.1.0 build:monolith
> env-cmd -f .env.monolith react-scripts build

Creating an optimized production build...
```

```
9af5f53e8f62: Layer already exists
bd51b4d942ca: Pushed
0bba9ace811c: Pushed
3.0.0: digest: sha256:3e37867a467ee2c22cc159c04ac94de51616daef034599ff63d9bf4236bdd4d5 size: 2841
DONE
--------------------
```